

QUICK REVISION GUIDE

Fundamentals of Algebraic Expressions & Polynomials

Class VII / Grade 7 Mathematics — Chapter 13

1. Key Building Blocks

Constant:

A symbol having a fixed numerical value. **Examples:** 3, -2 , $\frac{5}{9}$, 0.6, 5000.

Variable (Literal):

A symbol (usually an English letter) that can take various numerical values. **Examples:** x, y, z, p, q, s, m, n .

Algebraic Expression:

A combination of constants and variables connected by operational signs ($+$, $-$, $\times$, $\div$). **Example:** $3x + 4y - 2xy$.

Terms of an Expression:

Several parts of an algebraic expression separated by $+$ or $-$ signs.

- In $2x - 5y + 3xyz$, the terms are $2x$, $-5y$, and $3xyz$.
- In $a^2b - 2ab^2 + 5b^2a - 8$, the terms are a^2b , $-2ab^2$, $5b^2a$, and -8 .

Factors of a Term:

When quantities are multiplied to form a term, each is a factor.

- Numerical Factor (Constant factor):** In $-9a^2b$, the numerical factor is -9 .
- Literal Factor (Variable factor):** In $-9a^2b$, the literal factors are a, b, a^2, ab, a^2b .

Coefficients:

Any factor of a term is the coefficient of the product of the remaining factors.

- Numerical Coefficient:** The constant multiplier attached to the variables. (e.g., in $8x^2y$, it is 8; in $-2xyz^2$, it is -2).
- Literal Coefficient:** The non-numeric variable product. (e.g., in $-2xyz^2$, it is xyz^2).

Constant Term:

A term which has no literal factor (e.g., in $3x - 4y + 2$, the constant term is $+2$).

2. Types of Expressions & Terms

2.1 A. Classification by Number of Terms

Type	No. of Terms	Examples
Monomial	Exactly 1 term	$5x$, $-6z^2$, $\frac{13xyz}{7p}$, -29 , $\frac{43}{2}ab$
Binomial	Exactly 2 terms	$5x + 3z$, $2xy - \frac{3}{z}$, $a^2b - 3b^2c$
Trinomial	Exactly 3 terms	$x + 2y - 3z$, $z^2 - \frac{2x}{y} - 27$, $-3h^4 + 7h - 7$
Multinomial	More than 1 term	Any expression with 2, 3, 4 or more terms (includes binomials and trinomials).

2.2 B. Like vs. Unlike Terms

- **Like Terms:** Terms having the **exact same literal coefficients** (same variables with identical powers), regardless of numerical coefficients.
 - Examples: $3xy$ and $-5xy$; $6ab^2$ and $9b^2a$ (since $ab^2 = b^2a$).
- **Unlike Terms:** Terms that do not have identical literal coefficients.
 - Examples: $2a^2b$ and $3ab^2$ (exponents on a and b differ); $3x$ and $3y$.

3 3. Polynomials & Degree Rules

Definition of a Polynomial

An algebraic expression in which the variables involved have **only non-negative integer powers** ($0, 1, 2, 3, \dots$) is called a **polynomial**.

- **Valid Polynomial Examples:** $3x + 7$, $2y^2 - 5y + 1$, $\frac{2}{3}x^4 - 7x + 15$, $\frac{x}{\sqrt{2}} - \frac{1}{\sqrt{2}}$.
- **NOT Polynomials:**
 - $x + \frac{1}{x} = x + x^{-1}$ (power is -1 , a negative integer).
 - $8x^2 - 3x + 6\sqrt{x} + 1 = 8x^2 - 3x + 6x^{1/2} + 1$ (power is $\frac{1}{2}$, a fraction).
 - $5x^2 - \frac{2}{x} + 7$ (variable in denominator).

3.1 Degree of a Polynomial

1. **Polynomial in One Variable:** The **highest power** of the variable.
 - $3x + 7 \implies$ Degree 1
 - $2y^2 - 5y + 1 \implies$ Degree 2
 - $z^3 + 4z^2 + 3z - 6 \implies$ Degree 3
2. **Polynomial in Two or More Variables:** The **maximum sum of powers** of variables in any single term.
 - $x + y + xy \implies$ Term degrees: $1, 1, (1 + 1) = 2 \implies$ **Degree 2**
 - $a^2b + ab^2 + 3ab + 5 \implies$ Term degrees: $(2 + 1) = 3, (1 + 2) = 3, 2, 0 \implies$ **Degree 3**
 - $\frac{3}{4}a^3b^2c \implies$ Sum of powers $3 + 2 + 1 = 6 \implies$ **Degree 6**
3. **Constant Term / Number:** Any non-zero constant (like 15 or 7^5) has a degree of 0.

4 4. Operations on Algebraic Expressions

4.1 A. Addition of Polynomials

- Only **like terms** can be added together by summing their numerical coefficients.
- **Column Method:** Arrange polynomials in rows with like terms underneath each other.

Add $3x^2 + 5x - 4$, $-x^2 + 2x + 3$, $7x^2 - 3x + 8$:

$$\begin{array}{r}
 3x^2 + 5x - 4 \\
 -x^2 + 2x + 3 \\
 7x^2 - 3x + 8 \\
 \hline
 9x^2 + 4x + 7
 \end{array}$$

4.2 B. Subtraction of Polynomials

Golden Rule of Subtraction

To subtract an expression, **change the sign** ($+$ $\rightarrow$ $-$, $-$ $\rightarrow$ $+$) of every term in the subtrahend (the expression being subtracted), then add the like terms.

Subtract $3a + 4b - 2c$ from $6a - 2b + 3c$:

$$(6a - 2b + 3c) - (3a + 4b - 2c) = 6a - 2b + 3c - 3a - 4b + 2c = \mathbf{3a - 6b + 5c}$$

5. Top 5 Common Exam Pitfalls to Avoid!

CAUTION: Avoid These Common Mistakes

1. **Subtraction Order:** "Subtract A from B " means $B - A$, **NOT** $A - B$. Always write the expression after "from" first!
2. **Forgetting to Distribute Negative Sign:** When writing $B - (A)$, remember the negative sign flips the sign of **every single term** inside A .
3. **Constant Base Exponents:** The degree of 7^5 is 0, not 5, because $7^5 = 16807$ is a constant number with no variable!
4. **Simplify Before Classifying:** Always combine like terms first before classifying an expression. *Example:* $7x^2 - 3x^2 = 4x^2$, which is a **monomial**, not a binomial!
5. **Unlike Terms Exponents:** a^2b and ab^2 are **unlike terms** because the exponent 2 belongs to a in the first term, but to b in the second term.